Research Onion

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The Research Onion - Introduction

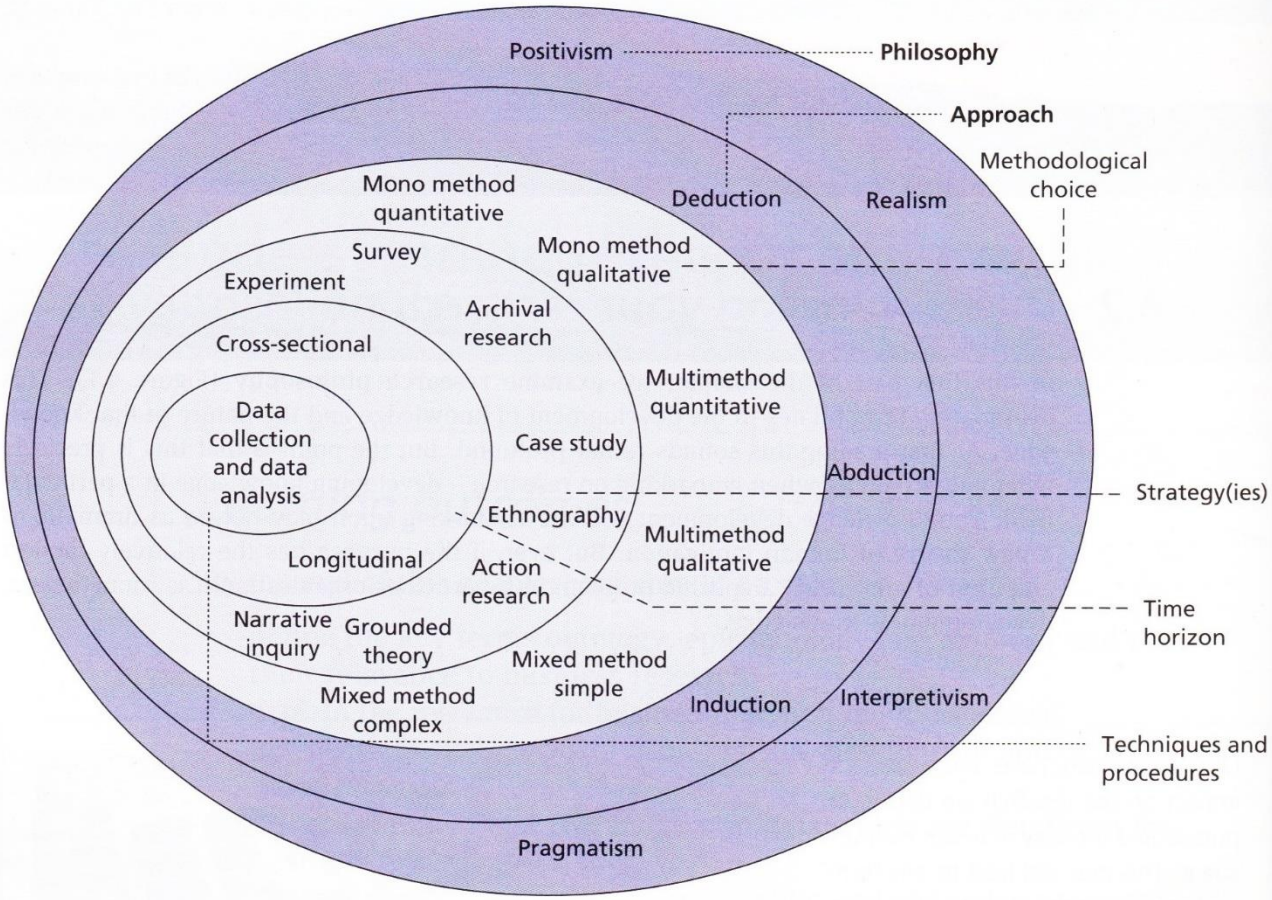
- Proposed by Saunders, Lewis and Thornhill [2000].

The Research Onion - Introduction

- This model symbolises the research design as an onion, consisting of various layers:
  - the research philosophy,
  - research approach,
  - research strategy,
  - time line
  - data collection.

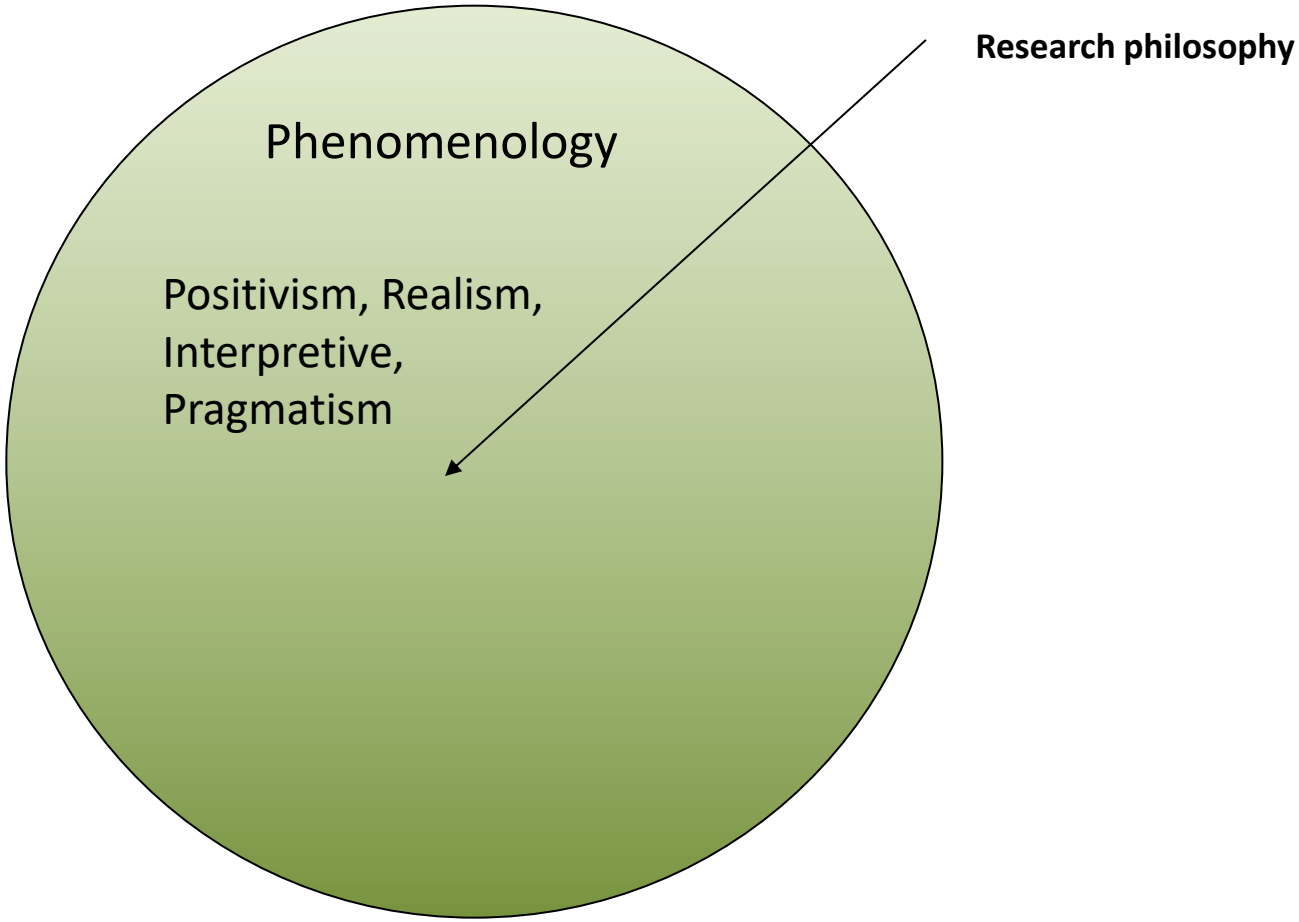
According to this model, the centre of the research design is the collection of data

The Research Onion – Introduction



Source: © Mark Saunders, Philip Lewis and Adrian Thornhill 2011

The Research Onion - Introduction

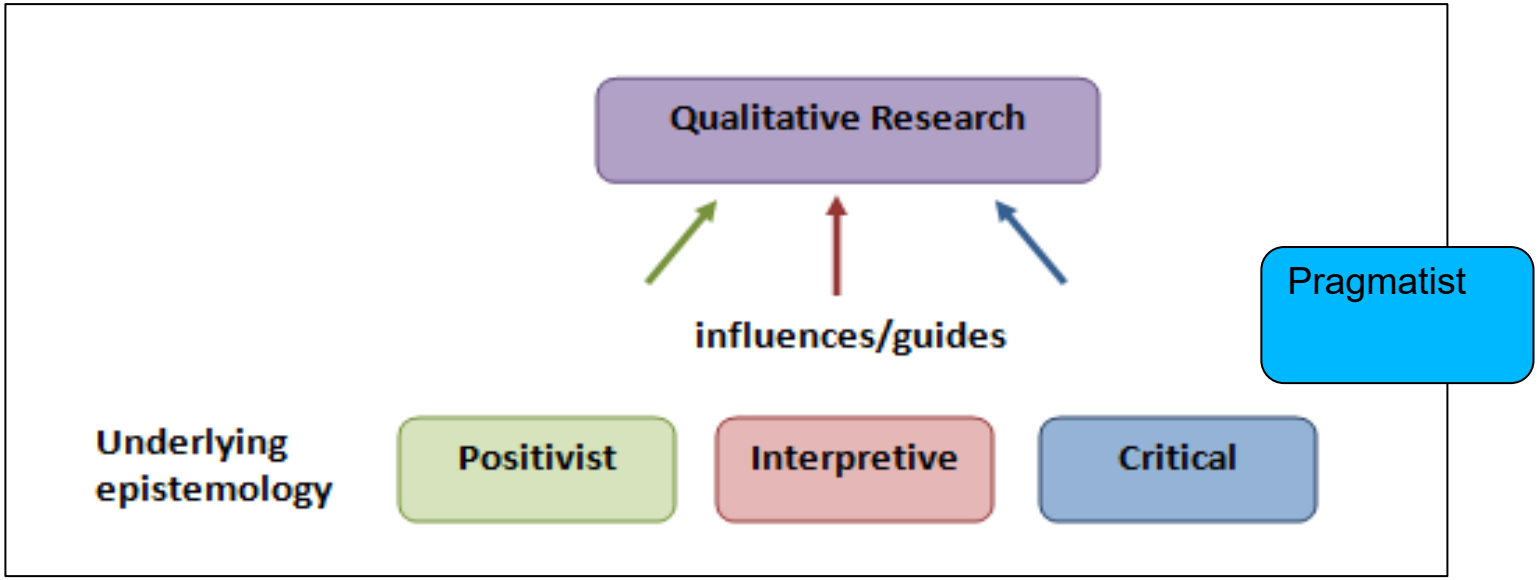


The Research Onion - Introduction

What is your Epistemological stance?

The Research Onion – Research Philosophy

All research (whether quantitative or qualitative) is based on some underlying assumptions about what constitutes 'valid' research and which research methods are appropriate.



The Research Onion – Research Philosophy

Positivists generally assume that reality is objectively given and can be described by measurable properties which are independent of the observer (researcher) and his or her instruments.

Positivist studies generally attempt to test theory, in an attempt to increase the predictive understanding of phenomena.

The Research Onion – Research Philosophy

Studies in information systems is referred to as positivist if there was evidence of formal propositions, quantifiable measures of variables, hypothesis testing, and the drawing of inferences about a phenomenon from the sample to a stated population.

Interpretive Research

Interpretive researchers start out with the assumption that access to reality (given or socially constructed) is only through social constructions such as language, consciousness and shared meanings.

The philosophical base of interpretive research is hermeneutics and phenomenology [Boland, 1985].

Interpretive Research

Interpretive studies generally attempt to understand phenomena through the meanings that people assign to them and interpretive methods of research in IS are ‘aimed at producing an understanding of the context of the information system, and the process whereby the information system influences and is influenced by the context’ [Walsham, 1993].

Critical research

Critical researchers assume that social reality is historically constituted and that it is produced and reproduced by people [Ngwenyama and Lee, 1997].

Although people can consciously act to change their social and economic circumstances, critical researchers recognise that their ability to do so is constrained by various forms of social, cultural and political domination.

Critical research

The main task of critical research is seen as being one of social critique, whereby the restrictive and alienating conditions of the status quo are brought to light. Critical research focuses on the oppositions, conflicts and contradictions in contemporary society, and seeks to emancipate i.e. it helps to eliminate the causes of alienation and domination.

Pragmatism

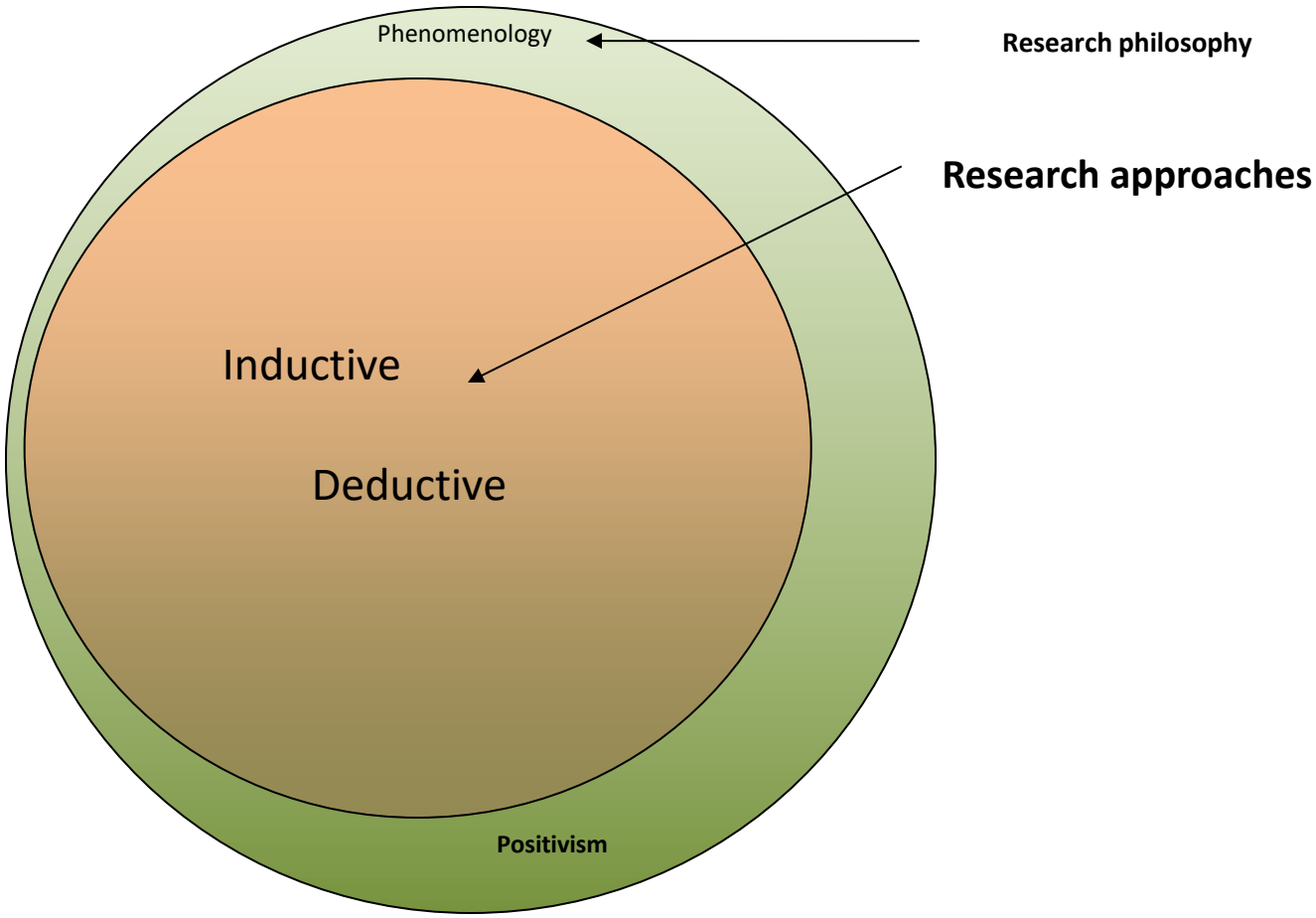
Definition of PRAGMATISM

***1*:** a practical approach to problems and affairs <tried to strike a balance between principles and *pragmatism*>

***2*:** an American movement in philosophy founded by C. S. Peirce and William James and marked by the doctrines that the meaning of conceptions is to be sought in their practical bearings, that the function of thought is to guide action, and that truth is preeminently to be tested by the practical consequences of belief

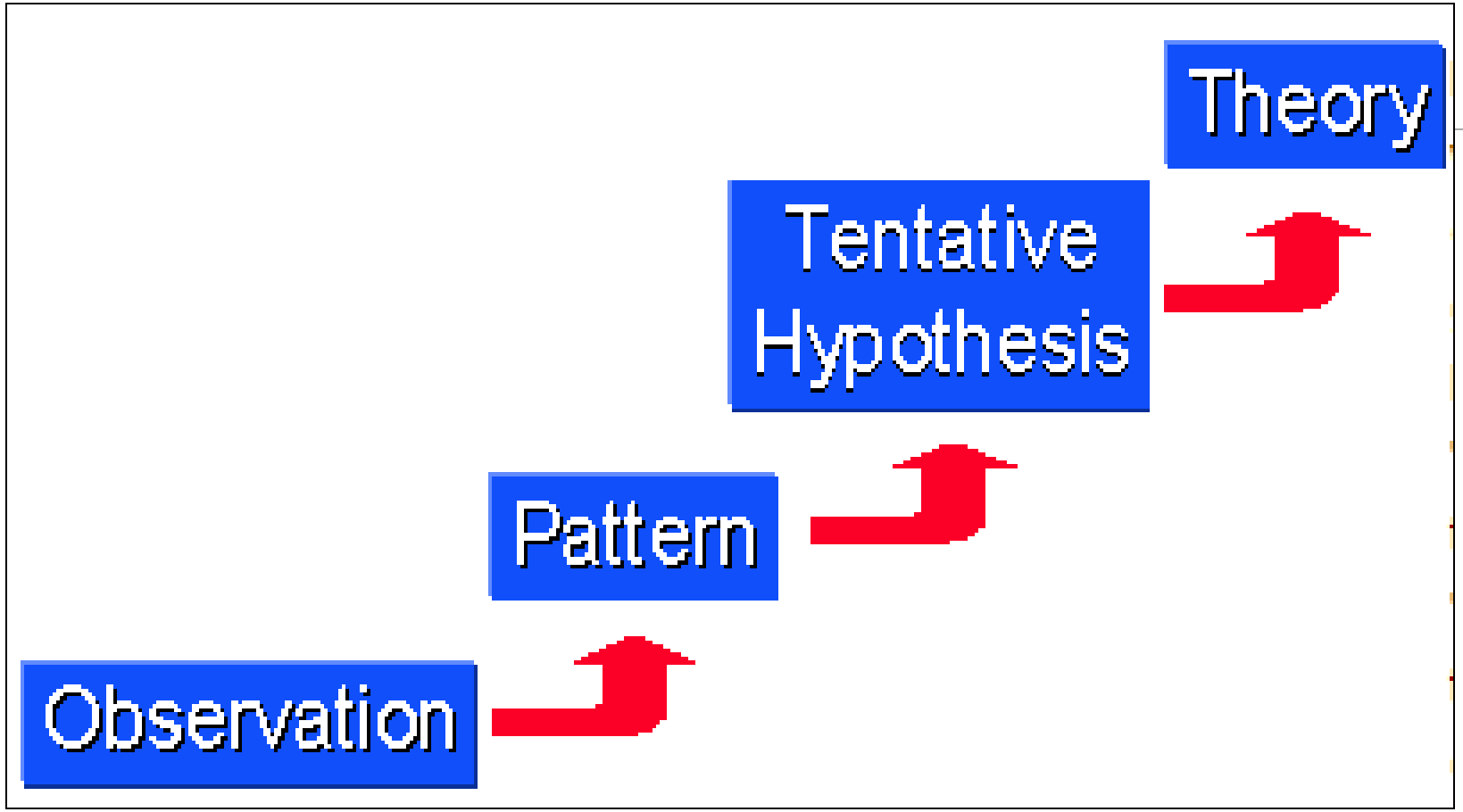
https://www.merriam-webster.com/dictionary/pragmatism

The Research Onion - Approach



Inductive Reasoning

In inductive reasoning, the research study begins with specific observations and measures, it detects patterns and regularities, formulates some tentative hypotheses (research questions/goals) that can be explored, and finally ends up developing some general conclusions or theories.



Trochim, 2008

Deductive Reasoning

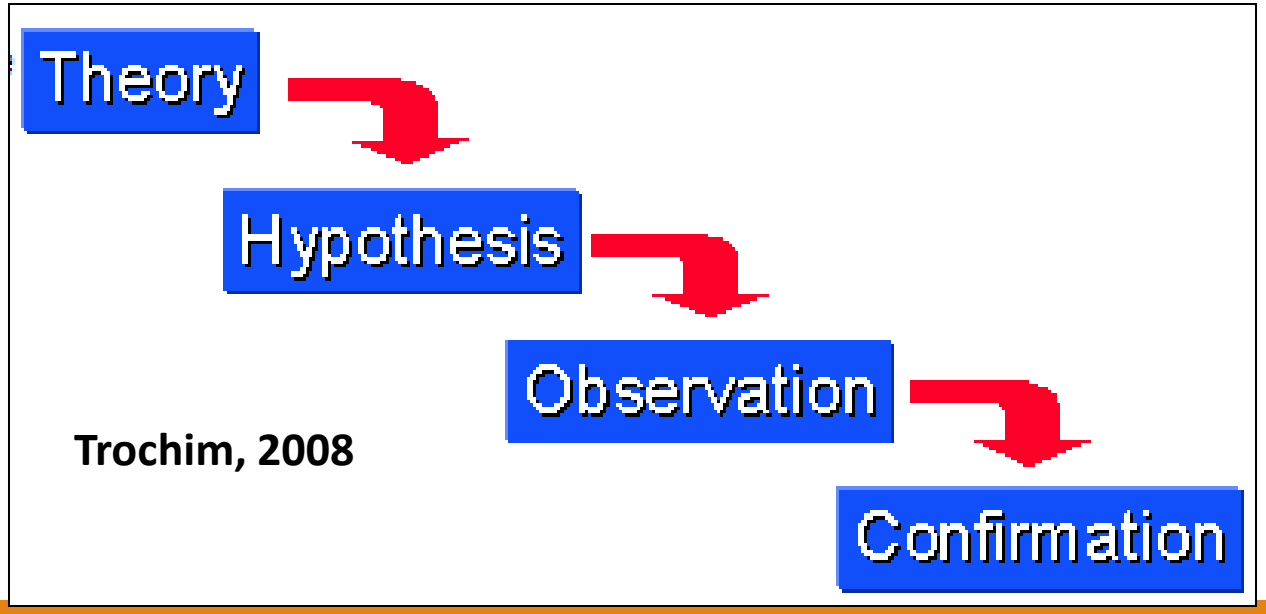
Deductive reasoning works from the more general to the more specific [Saunders, Lewis and Thornhill, 2000].

This is often referred to as a ‘top-down’ approach. The research study may begin with a *theory* about a topic of interest.

It is then narrowed down into more specific *hypotheses (research questions/goals)* that can be tested.

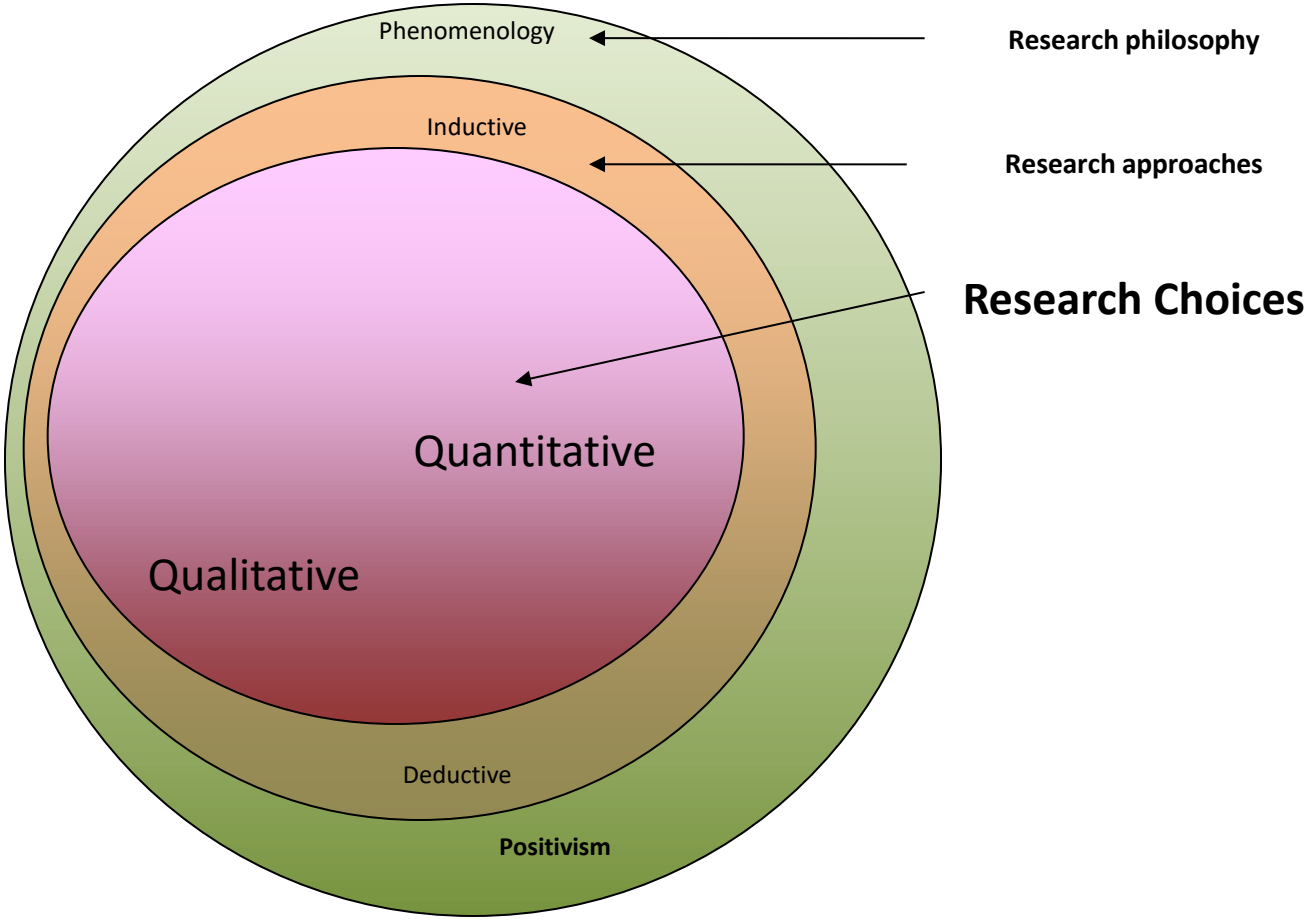
Deductive Reasoning

It is then narrowed down even further to collect *observations* to address the hypotheses. This ultimately leads to the test of the hypotheses with specific data - a *confirmation* (or not) of the original theories.



Trochim, 2008

The Research Onion – Research strategies



Research Choices

Choices: Quantitative vs. qualitative

■ **Quantitative research:**

♦ focuses on verifying hypotheses (deductive) or finding patterns (inductive) using typically *large amounts of data*

■ **Qualitative research:**

♦ focuses on understanding the important characteristics of typically *small samples of data*

■ **Example:** investigate users' response to an interface

♦ quantitative approach: collect ratings, verify user acceptance

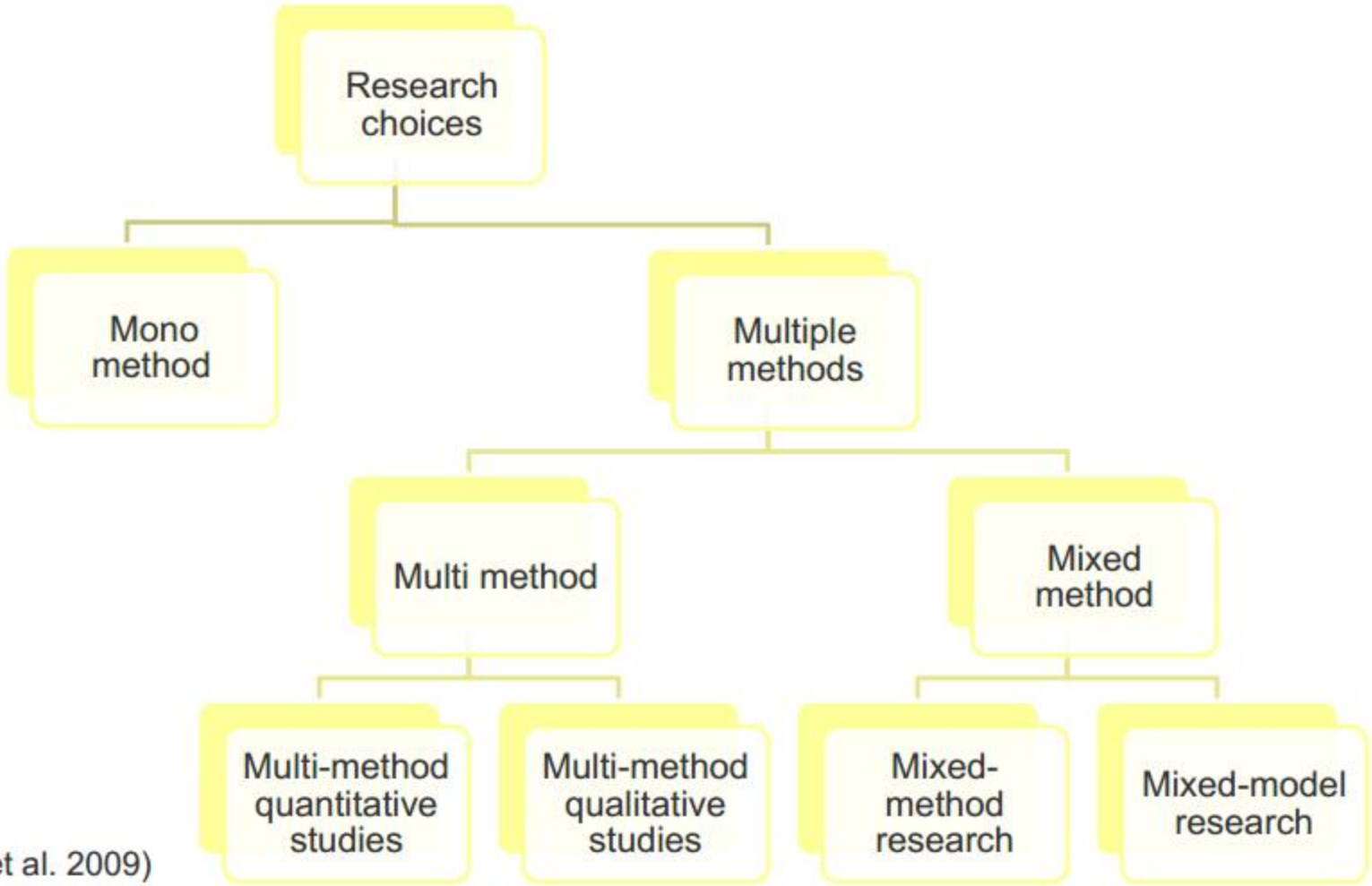
♦ qualitative approach: understand *why* users interact with the interface in certain ways

Quants/Qual or both?

Indeed, some writers, such as Guba and Lincoln (1994:105), argue that questions of research methods are of secondary importance to questions of which paradigm is applicable to your research.

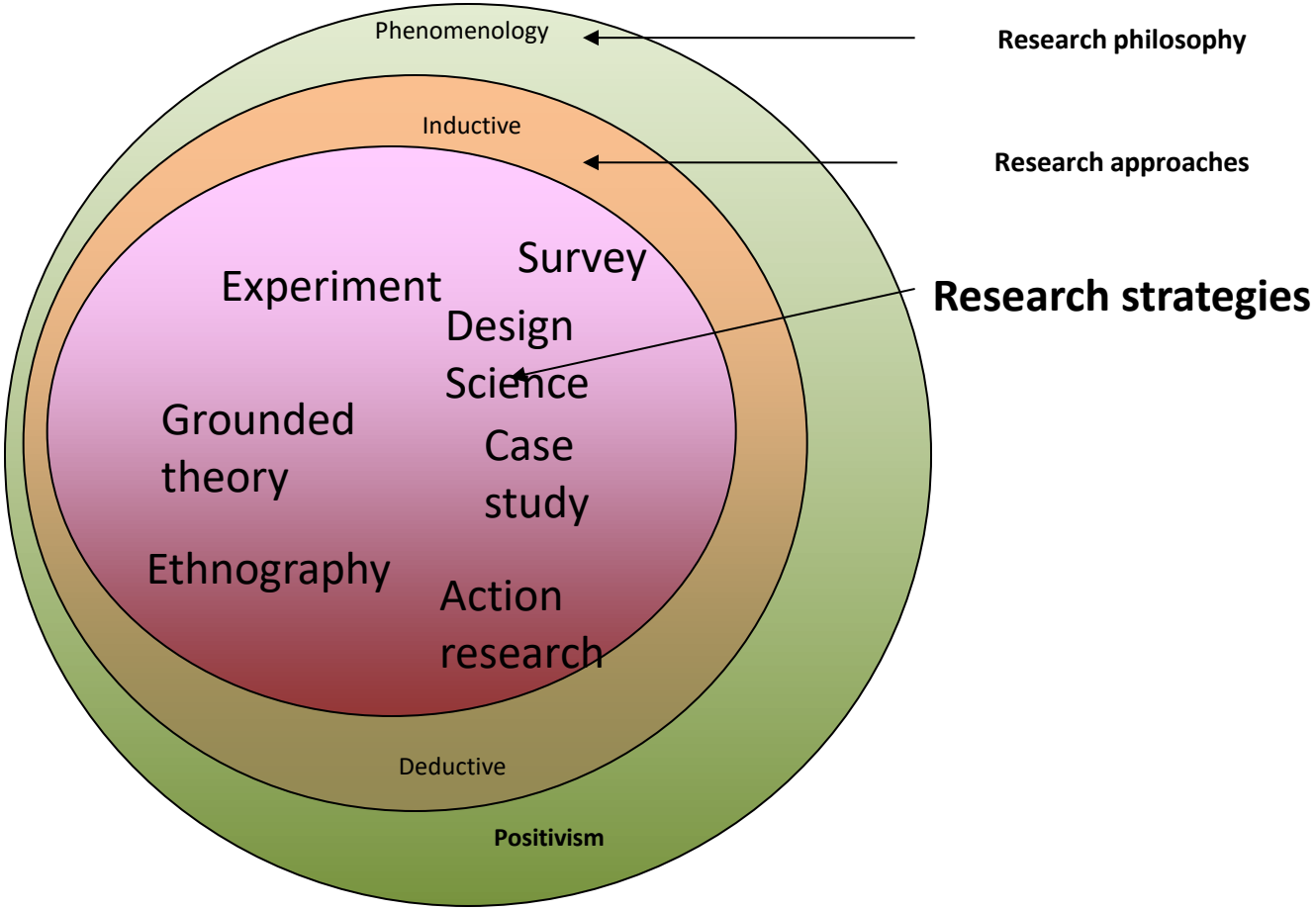
They note: both qualitative and quantitative methods may be used appropriately with any research paradigm.

Research Choices



(Saunders et al. 2009)

The Research Onion – Research strategies

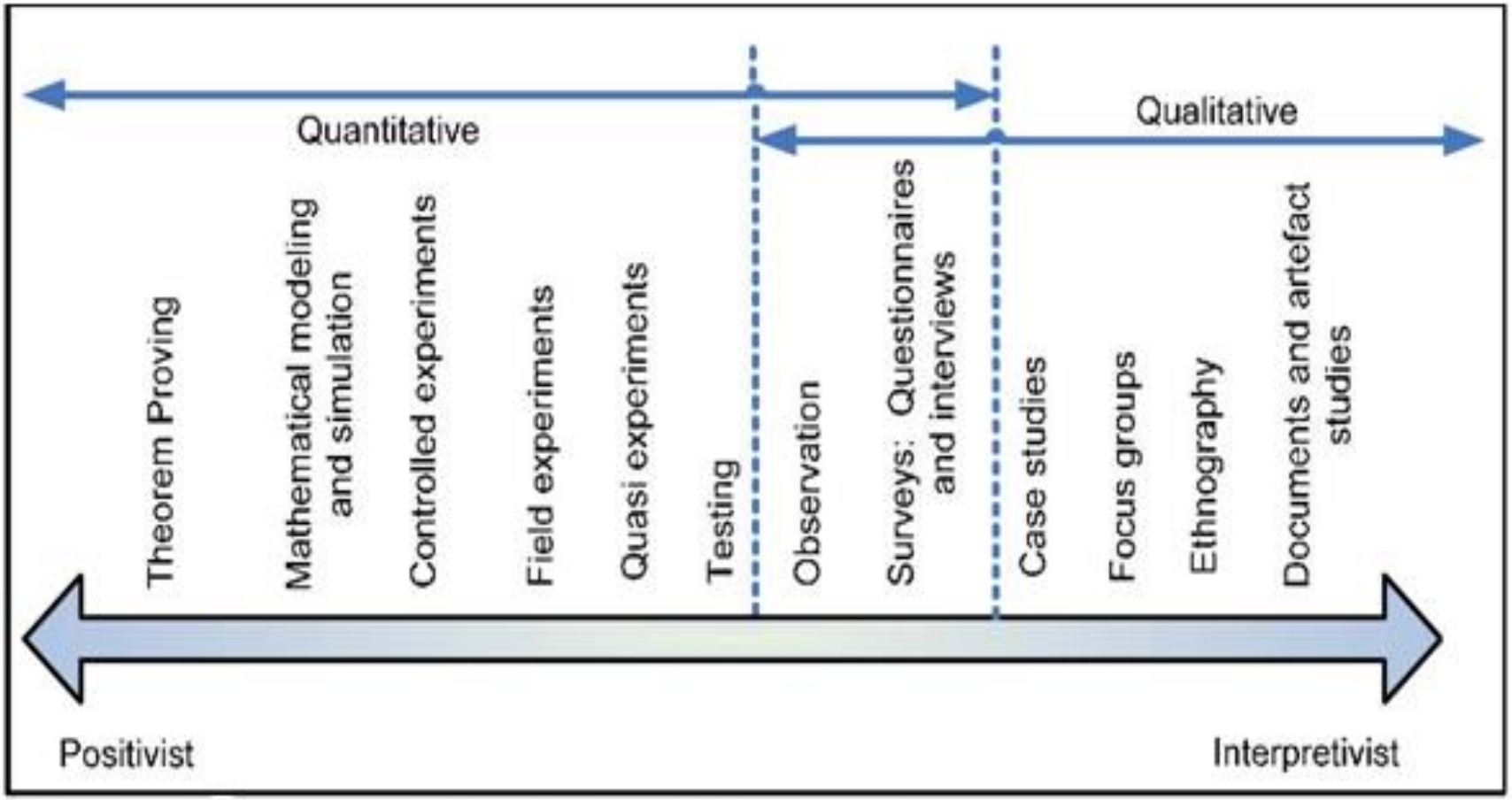


Research Strategy

A research strategy is a method of inquiry which moves from the underlying philosophical assumptions to research design and data collection.

The choice of research method influences the way in which the researcher collects data [Meyers, 2004]. Specific research methods also imply different skills, assumptions and research practices.

Research Strategy



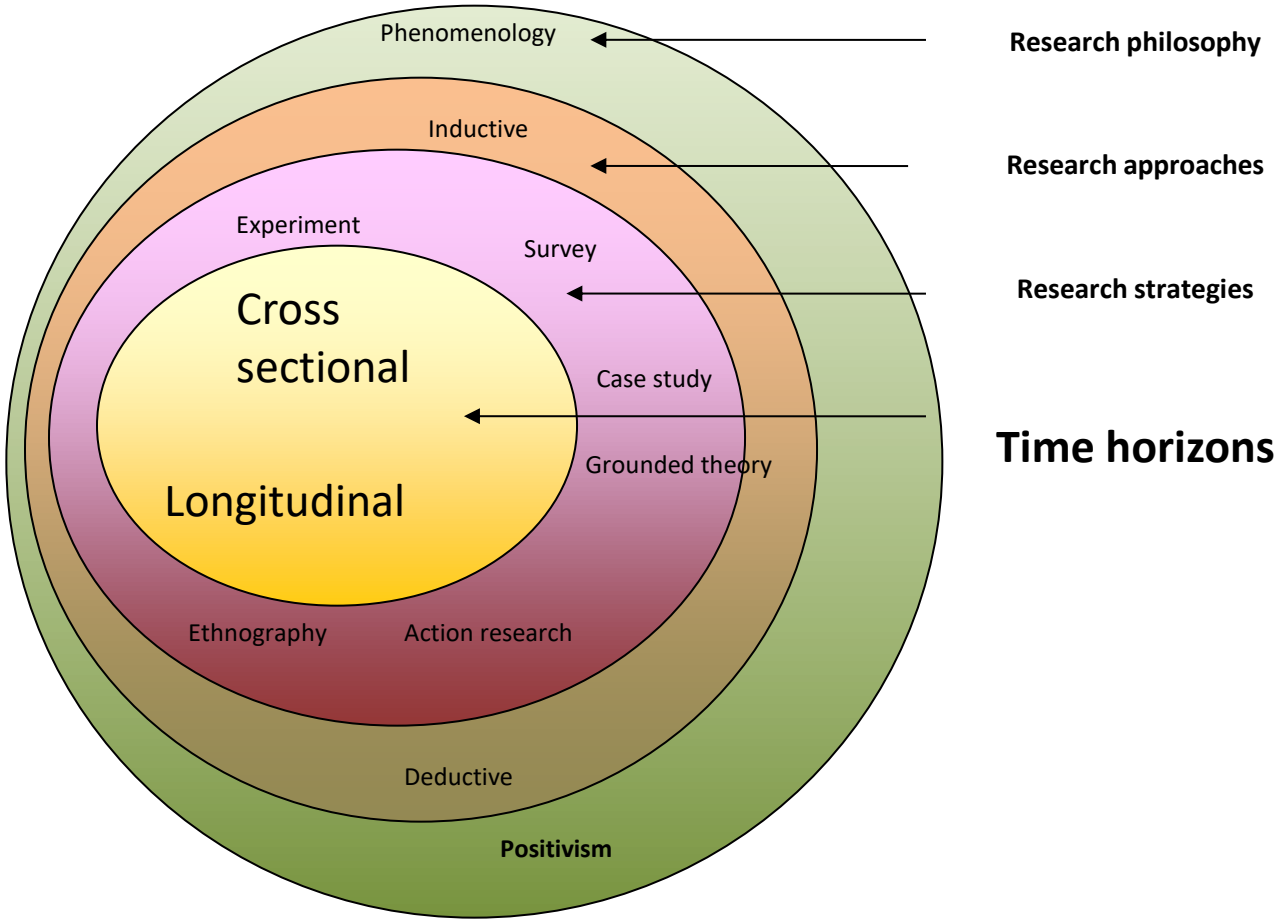
Research Strategy

Research method	Research method description
Appreciative enquiry	Researcher acts as facilitator with technique that can be used to improve wider range of contexts. • Discovery – an 'appreciation' of what already exists. • Dream – what can be. • Design – formulates vision and strategy. • Delivery – implement plans
Action research	Iterative method for determining current situation of interest and then designing an intervention. Researcher collaborates with practitioners and deliberately intervenes. Contributes to both research and practice
Case study	Explores a single entity or phenomenon bounded by time and activity. Collects detailed information through variety of data collection methods over sustained period of time.
Ethnography and participant observation	Researcher studies an intact group of individuals in a natural setting over a specific period of time. Observes what people are doing as well as what they say they are doing – participant observer.
Grounded theory	Seeks to develop theory that is grounded in data systematically gathered and analysed. Rigorous and detailed method.
Hermeneutics	Theory of interpretation of meaning. Primarily concerned with the meaning of text or text-analogue.
Semiotics	Study of signs. Inherently interpretive of nature.

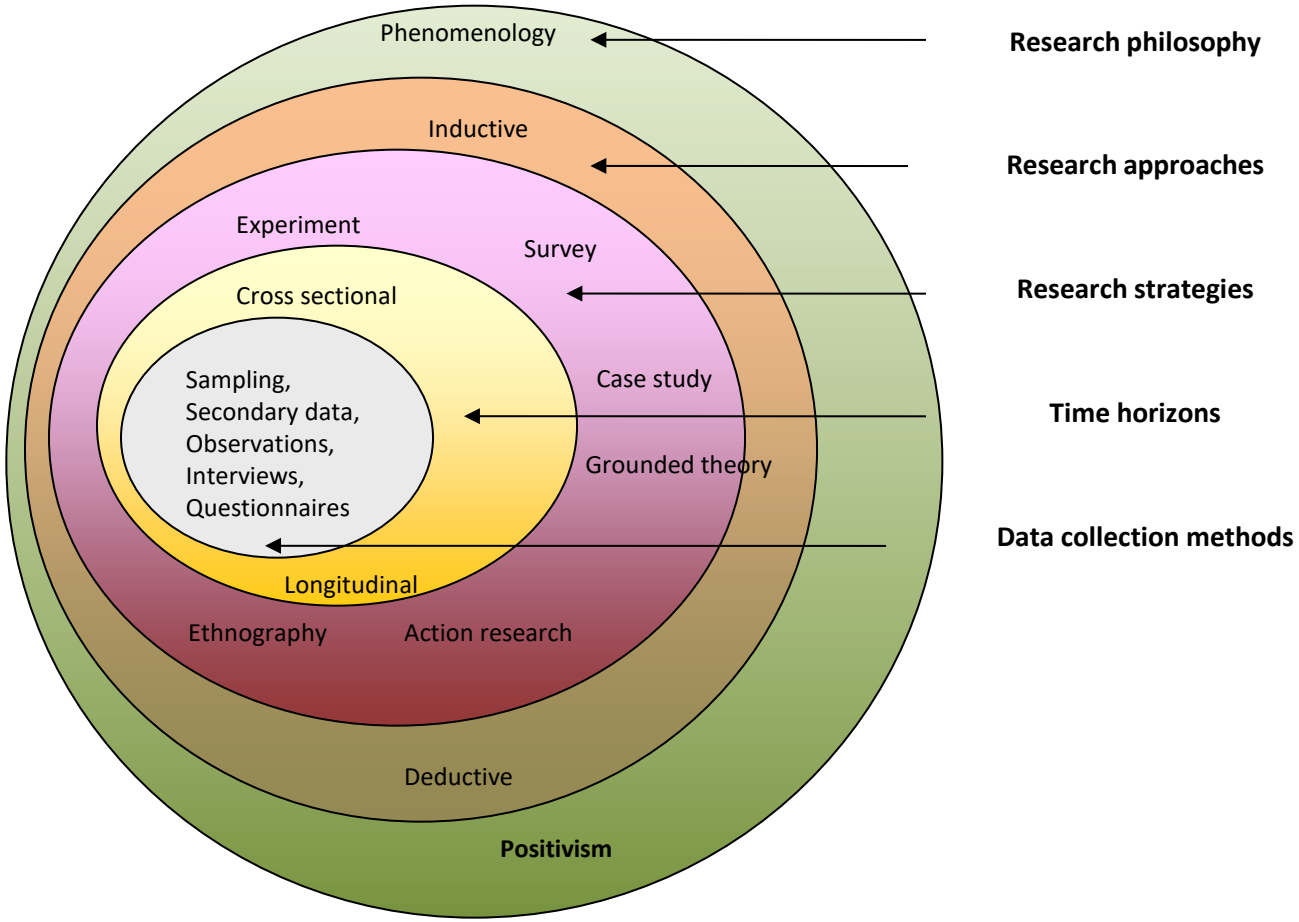
Selection of Research Strategy

Approach	Characteristics	Research question 1	Research question 2	Research question 3
Action research	Focus on what practitioners do	✓	✓	
	Explicit criteria			
	Practitioners and researchers with mutual goals			
	Apply theory with goal to enhance			
Case study	Investigator has little control	✓	✓	✓
	Contemporary phenomenon with real-life context	✓	✓	✓
	Explores a single entity or phenomenon bounded by time and activity		✓	✓
	Study life cycles	✓	✓	✓
Content analysis	Study of written documentation or texts			
	Study of authorship, authenticity and meaning			
	Explores undefined number of variables			
Ethnographic research	Active participation			
	Observational data		✓	✓
	Social contact with participants			
	Extended depth study			I
	Limited to one field study			
Experiments	To study the effect of an influence on a carefully controlled sample			
	Observational measures of dependent variable			
	Manipulation of variable to determine level and kind			
Grounded theory	Start with a phenomena			
	Data sampling with perspective			
	Theoretical account of the general features	✓		✓
	Generation of theories of process, sequence, and change pertaining to organisations, positions, and social interaction		✓	

The Research Onion – Time Horizons



The Research Onion – Data Collection



Design Science Research: Link to Slides

Create your own design

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https://docs.google.com/document/d/1uQHq40GxYT9KuHIJwtoqhLhIYHNk8CqN/edit?pli=1

10 minutes

Load your own table to the shared Excel Spreadsheet.